MOHAMMED
SAAD

2026

PROJECT

DEVOPS

DEVOPS

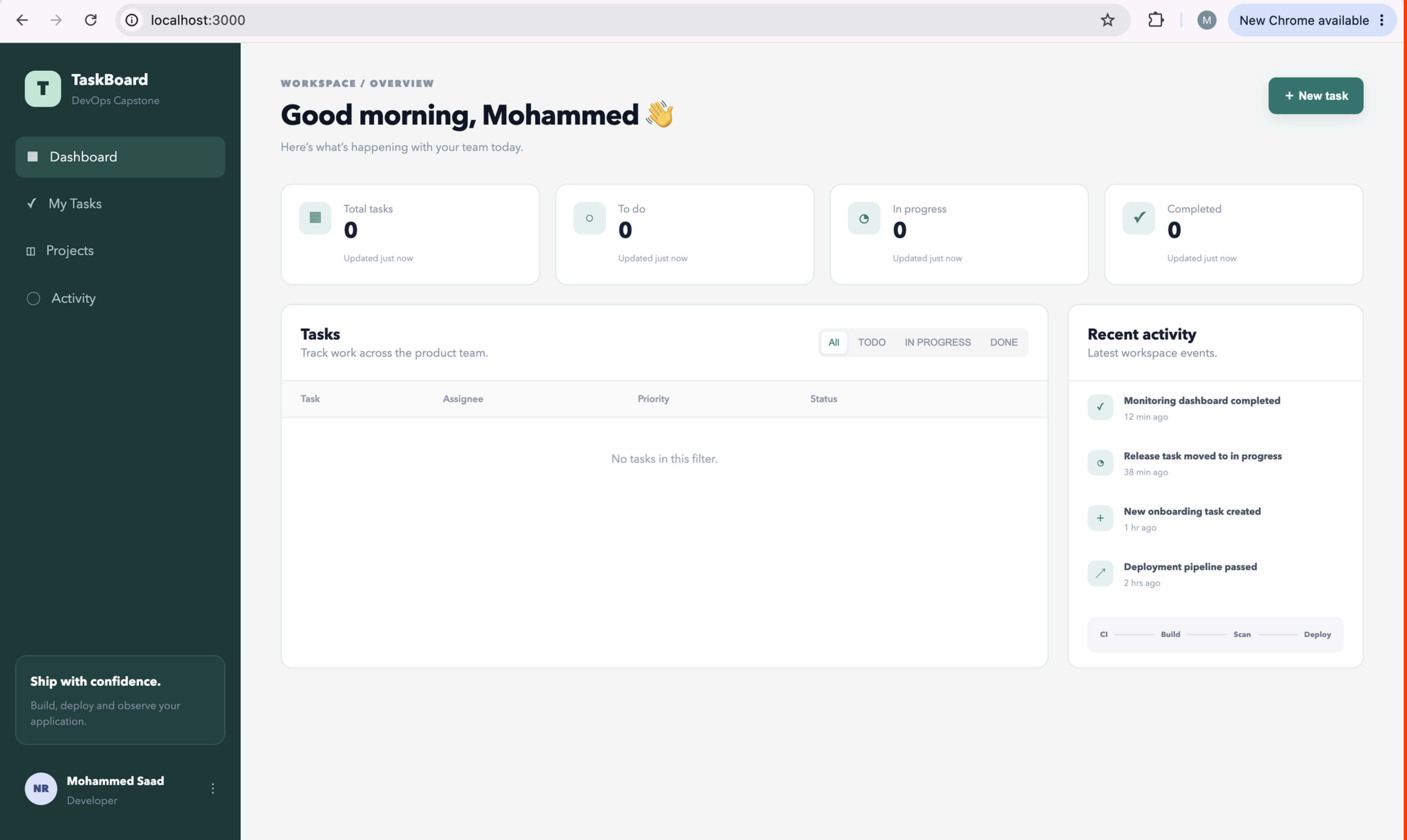
Capstone

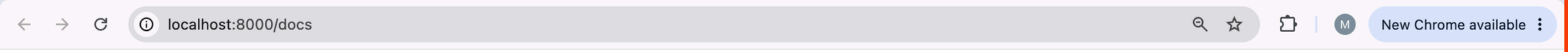
DEVOPS

PROJECT



A TaskBoard dashboard screenshot on the right, with a subtle background of a deployment pipeline or cloud infrastructure


localhost:3000
New Chrome available
TaskBoard
DevOps Capstone
Dashboard
My Tasks
Projects
Activity
Ship with confidence.
Build, deploy and observe your application.
Mohammed Saad
Developer
WORKSPACE / OVERVIEW
Good morning, Mohammed 👋
Here's what's happening with your team today.
+ New task
Total tasks
0
Updated just now
To do
0
Updated just now
In progress
0
Updated just now
Completed
0
Updated just now
Tasks
Track work across the product team.
All
TODO
IN PROGRESS
DONE
Task
Assignee
Priority
Status
No tasks in this filter.
Recent activity
Latest workspace events.
Monitoring dashboard completed
12 min ago
Release task moved to in progress
38 min ago
New onboarding task created
1 hr ago
Deployment pipeline passed
2 hrs ago
CI
Build
Scan
Deploy



TaskBoard API 1.0.0 OAS 3.1

/openapi.json

default

- GET /metrics Metrics
- GET / Root
- GET /health Health
- GET /ready Ready
- GET /api/tasks List Tasks
- POST /api/tasks Create Task
- GET /api/tasks/stats Stats
- GET /api/tasks/{task_id} Get Task
- PUT /api/tasks/{task_id} Update Task
- DELETE /api/tasks/{task_id} Delete Task

Schemas

- HTTPValidationError > Expand all object
- StatsOut > Expand all object
- TaskCreate > Expand all object
- TaskOut > Expand all object
- TaskUpdate > Expand all object
- ValidationError > Expand all object

2026

```
mohammedsaad@Saads-MacBook-Air 21-final_project % git rev-parse HEAD
0d313ce0f322e352dd00823e9e75e5eaebebcd48
```

DATABASE GIT

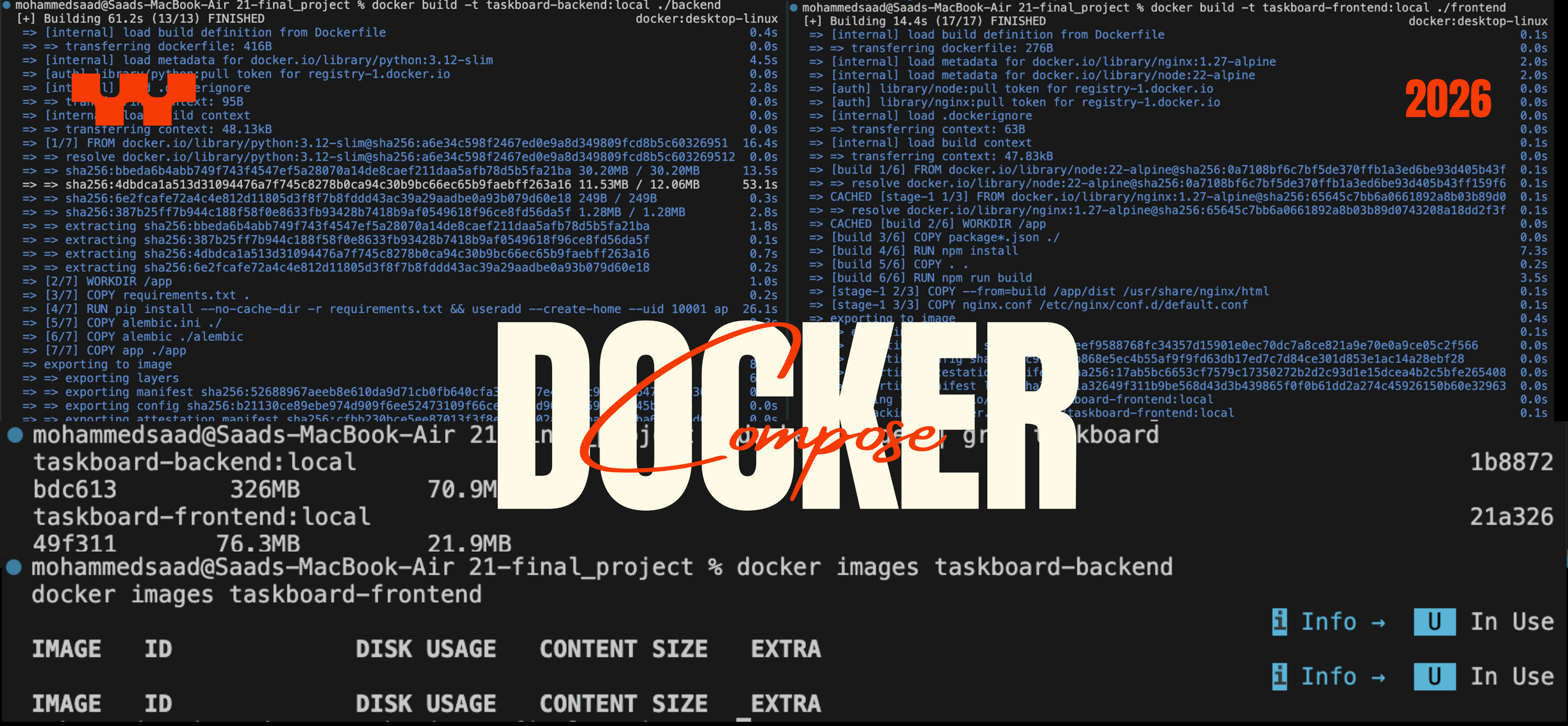
Make a small, visible application change (for example, edit a dashboard label), then commit it with a meaningful message:

git status

git add frontend/src/main.jsx

git commit -m "chore: refine TaskBoard dashboard label"

git log --oneline -10



The frontend image uses a Node build stage and a compact Nginx runtime stage.

This output confirms that taskboard-backend:local and taskboard-frontend:local were produced.

SECURITY

2026

Scan both images with Trivy, focusing on HIGH and CRITICAL findings:

trivy image --severity HIGH,CRITICAL taskboard-backend:local

trivy image --severity HIGH,CRITICAL taskboard-frontend:local

```
mohammedsaad@Saads-MacBook-Air 21-final_project % trivy image --severity HIGH,CRITICAL taskboard-backend:loc
al
trivy image --severity HIGH,CRITICAL taskboard-frontend:local
2026-10-16T19:01:33+05:30       INFO    [vulndb] Need to update DB
2026-10-16T19:01:33+05:30       INFO    [vulndb] Downloading vulnerability DB...
2026-10-16T19:01:33+05:30       INFO    [vulndb] Downloading artifact...        repo="mirror.gcr.io/aquasec/
trivy-db:2"
128.03 MiB / 128.03 MiB [-------------------------------------------------------] 100.00% 3.05 MiB p/s 42s
2026-10-16T19:02:17+05:30       INFO    [vulndb] Artifact successfully downloaded       repo="mirror.gcr.io/
aquasec/trivy-db:2"
2026-10-16T19:02:17+05:30       INFO    [vuln] Vulnerability scanning is enabled
2026-10-16T19:02:17+05:30       INFO    [secret] Secret scanning is enabled
2026-10-16T19:02:17+05:30       INFO    [secret] If your scanning is slow, please try '--scanners vuln' to d
isable secret scanning
2026-10-16T19:02:17+05:30       INFO    [secret] Please see https://trivy.dev/docs/v0.76/guide/scanner/secre
t#recommendation for faster secret detection
2026-10-16T19:02:20+05:30       INFO    [python] Licenses acquired from one or more METADATA files may be su
bject to additional terms. Use `--debug` flag to see all affected packages.
2026-10-16T19:02:21+05:30       INFO    Detected OS     family="debian" version="13.7"
2026-10-16T19:02:21+05:30       INFO    [debian] Detecting vulnerabilities...   os_version="13" pkg_num=87
2026-10-16T19:02:21+05:30       INFO    Number of language-specific files       num=1
2026-10-16T19:02:21+05:30       INFO    [python-pkg] Detecting vulnerabilities...
2026-10-16T19:02:21+05:30       WARN    Using severities from other vendors for some vulnerabilities. Read h
ttps://trivy.dev/docs/v0.76/guide/scanner/vulnerability#severity-selection for details.
2026-10-16T19:02:21+05:30       INFO    Table result includes only package filenames. Use '--format json' op
tion to get the full path to the package file.

Report Summary

┌──────────────────────────────────────────────────────────────────────────────────────────┬────────────┬──────────────┬─────────┐
│                                          Target                                          │    Type    │ Vulnerabil   │
ities  │ Secrets │
├──────────────────────────────────────────────────────────────────────────────────────────┼────────────┼──────────────┼─────────┤
│ taskboard-backend:local (debian 13.7)                                                    │   debian   │      44      │
-      │
├──────────────────────────────────────────────────────────────────────────────────────────┼────────────┼──────────────┼─────────┤
│ usr/local/lib/python3.12/site-packages/SQLAlchemy-2.0.36.dist-info/METADATA              │ python-pkg │      0       │
-      │
├──────────────────────────────────────────────────────────────────────────────────────────┼────────────┼──────────────┼─────────┤
│ usr/local/lib/python3.12/site-packages/alembic-1.14.0.dist-info/METADATA                 │ python-pkg │      0       │
-      │
├──────────────────────────────────────────────────────────────────────────────────────────┼────────────┼──────────────┼─────────┤
│ usr/local/lib/python3.12/site-packages/annotated_types-0.8.0.dist-info/METADATA          │ python-pkg │      0       │
-      │
├──────────────────────────────────────────────────────────────────────────────────────────┼────────────┼──────────────┼─────────┤
```

The captured report shows that Trivy scanned the image OS and Python packages. Treat reported HIGH/CRITICAL findings as remediation work: update the base image or affected package, rebuild, and scan again. A scan is valuable because it reports risk; it is not automatically a clean result.

2026

```
mohammedsaad@Saads-MacBook-Air terraform % terraform init
Initializing the backend...

Initializing modules...

Initializing provider plugins...
- Finding hashicorp/cloudinit versions matching ">= 2.0.0"...
- Finding hashicorp/aws versions matching ">= 4.33.0, ~> 5.0, >= 5.30.0, >= 5.95.0, < 6.0.0"...
- Finding hashicorp/time versions matching ">= 0.9.0"...
- Finding hashicorp/tls versions matching ">= 3.0.0"...
- Finding hashicorp/null versions matching ">= 3.0.0"...
- Installing hashicorp/cloudinit v2.4.1...
- Installed hashicorp/cloudinit v2.4.1 (signed by HashiCorp)
- Installing hashicorp/aws v5.100.0...
- Installed hashicorp/aws v5.100.0 (signed by HashiCorp)
- Installing hashicorp/time v0.14.2...
- Installed hashicorp/time v0.14.2 (signed by HashiCorp)
- Installing hashicorp/tls v4.4.1...
- Installed hashicorp/tls v4.4.1 (signed by HashiCorp)
- Installing hashicorp/null v3.3.2...
- Installed hashicorp/null v3.3.2 (signed by HashiCorp)

Terraform has created a lock file .terraform.lock.hcl to record the provider
selections it made above. Include this file in your version control repository
so that Terraform can guarantee to make the same selections by default when
you run "terraform init" in the future.

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
```

```
mohammedsaad@Saads-MacBook-Air terraform % terraform fmt
terraform validate
Success! The configuration is valid.

mohammedsaad@Saads-MacBook-Air terraform %
```

TERRAFORM

Show the AWS infrastructure code in terraform/, then validate and plan it. Do not place credentials in the repository.

cd terraform
terraform init
terraform fmt -recursive
terraform validate
terraform plan

```
mohammedsaad@Saads-MacBook-Air 21-final_project % kubectl apply -f k8s/namespace.yaml
namespace/taskboard created
mohammedsaad@Saads-MacBook-Air 21-final_project % kubectl get namespace taskboard
NAME        STATUS   AGE
taskboard   Active   8s
mohammedsaad@Saads-MacBook-Air 21-final_project % kubectl get svc -n taskboard
kubectl get endpoints -n taskboard
NAME                        TYPE        CLUSTER-IP      EXTERNAL-IP   PORT(S)    AGE
taskboard-frontend          ClusterIP   10.110.28.164   <none>        80/TCP     29m
taskboard-postgres          ClusterIP   10.110.15.143   <none>        5432/TCP   29m
taskboard-taskboard-backend ClusterIP   10.102.141.53   <none>        8000/TCP   29m

NAME                        ENDPOINTS          AGE
taskboard-frontend                             29m
taskboard-postgres          10.244.0.80:5432   29m
taskboard-taskboard-backend                    29m
```

```
mohammedsaad@Saads-MacBook-Air 21-final_project % docker ps
CONTAINER ID   IMAGE                                  COMMAND                  CREATED       STATUS
PORTS                                                
```

```
mohammedsaad@Saads-MacBook-Air 21-final_project % helm template taskboard helm/taskboard
# Source: taskboard/templates/postgres.yaml
apiVersion: apps/v1
kind: Deployment
metadat  {na  : t  kboard-postgres}
spec:
  replicas
  selector: {matchLabels: {app: taskboard-postgres}}
  template:
    metadata: {labels: {app: taskboard-postgres}}
    spec:
      containers:
      - name: postgres
        image: postgres:16-alpine
        ports: [{containerPort: 5432}]
        env:
        - {name: POSTGRES_DB, value: "taskboard"}
        - {name: POSTGRES_USER, valueFrom: {secretKeyRef: {name: taskboard-postgres, key: username}}}
        - {name: POSTGRES_PASSWORD, valueFrom: {secretKeyRef: {name: taskboard-postgres, key: password
        readinessProbe: {exec: {command: ["sh","-c","pg_isready -U $POSTGRES_USER -d $POSTGRES_DB"]},
periodSeconds: 10}
        volumeMounts: [{name: data, mountPath: /var/lib/postgresql/data}]
      volumes: [{name: data, persistentVolumeClaim: {claimName: taskboard-postgres-data}}]

---
# Source: taskboard/templates/hpa.yaml
apiVersion: autoscaling/v2
kind: HorizontalPodAutoscaler
metadata: {name: taskboard-backend}
spec:
  scaleTargetRef: {apiVersion: apps/v1, kind: Deployment, name: taskboard-taskboard-backend}
  minReplicas: 2
  maxReplicas: 6
  metrics: [{type: Resource, resource: {name: cpu, target: {type: Utilization, averageUtilization: 60}

---
# Source: taskboard/templates/servicemonitor.yaml
apiVersion: monitoring.coreos.com/v1
kind: ServiceMonitor
metadata:
  name: taskboard-taskboard-backend
  labels:
    release: kube-prometheus-stack
spec:
  selector:
    matchLabels: {app: taskboard-backend}
  endpoints:
    - port: http
      path: /metrics
      interval: 15s
```

2026

Render the chart before installing it, then show the release:
helm template taskboard helm/taskboard
helm upgrade --install taskboard helm/taskboard -n taskboard --create-namespace
helm list -n taskboard

The rendered output includes the PostgreSQL Deployment, HorizontalPodAutoscaler, and ServiceMonitor. Use helm list -n taskboard live to show the installed release and revision.

HELM

```
mohammedsaad@Saads-MacBook-Air 21-final_project % grep -n -i -A 12 -B 4 ingress helm/taskboard/values-dev.yaml
1-replicaCount: 1
2:ingress:
3-  enabled: true
4-  host: taskboard.local
5-hpa:
6-  enabled: false
```

```
mohammedsaad@Saads-MacBook-Air 21-final_project % minikube addons enable ingress
kubectl get pods -n ingress-nginx
💡  ingress is an addon maintained by Kubernetes. For any concerns contact minikube on GitHub.
You can view the list of minikube maintainers at: https://github.com/kubernetes/minikube/blob/master/OWNERS
💡  After the addon is enabled, please run "minikube tunnel" and your ingress resources would be available at "127.0.0.1"
    ▪ Using image registry.k8s.io/ingress-nginx/controller:v1.15.1
    ▪ Using image registry.k8s.io/ingress-nginx/kube-webhook-certgen:v1.6.9
    ▪ Using image registry.k8s.io/ingress-nginx/kube-webhook-certgen:v1.6.9
🔎  Verifying ingress addon...
🌟  The 'ingress' addon is enabled
NAME                                        READY   STATUS      RESTARTS       AGE
ingress-nginx-admission-create-m2vtv        0/1     Completed   0              26d
ingress-nginx-admission-patch-vbqgp         0/1     Completed   1 (26d ago)    26d
ingress-nginx-controller-d7cd8c989-pvvgv    0/1     Unknown     19 (61m ago)   26d
```

INGRESS

Enable the ingress controller, deploy the Ingress, add the host mapping required by the local cluster, and open the application through the configured hostname.
minikube addons enable ingress
kubectl get ingress -n taskboard
minikube ip

For a local host such as taskboard.local, add the reported Minikube IP to /etc/hosts, then browse to http://taskboard.local.

This capture records an unhealthy controller (0/1 and Unknown). It belongs to the failure/troubleshooting evidence; wait for the controller to be Ready before demonstrating the hostname.

2026

```
mohammedsaad@Saads-MacBook-Air 21-final_project % kubectl get hpa -n taskboard
kubectl describe hpa -n taskboard
kubectl top pods -n taskboard
kubectl top nodes
NAME                REFERENCE                                  TARGETS              MINPODS   MAXPODS
taskboard-backend   Deployment/taskboard-taskboard-backend     cpu: <unknown>/60%   2         6
Name:                                                  taskboard-backend
Namespace:                                             taskboard
Labels:                                                app.kubernetes.io/managed-by=Helm
Annotations:                                           meta.helm.sh/release-name: taskboard
                                                       meta.helm.sh/release-namespace: taskboard
CreationTimestamp:                                     Fri, 16 Oct 2026 20:02:46 +0530
Reference:                                             Deployment/taskboard-taskboard-backend
Metrics:                                               ( current / target )
  resource cpu on pods  (as a percentage of request):  <unknown> / 60%
Min replicas:                                          2
Max replicas:                                          6
Deployment pods:                                       2 current / 0 desired
Conditions:
  Type           Status  Reason                   Message
  ----           ------  ------                   -------
  AbleToScale    True    SucceededGetScale        the HPA controller was able to get the target's cur
  ScalingActive  False   FailedGetResourceMetric  the HPA was unable to compute the replica count: fa
etch metrics from resource metrics API: the server is currently unable to handle the request (get poc
Events:
  Type     Reason                        Age                From                       Message
  ----     ------                        ----               ----                       -------
mohammedsaad@Saads-MacBook-Air 21-final_project % minikube addons list
minikube addons enable metrics-server
kubectl get pods -n kube-system | grep metrics-server
| metrics-server               | minikube | enabled ✅ | Kubernetes
💡  metrics-server is an addon maintained by Kubernetes. For any concer
You can view the list of minikube maintainers at: https://github.com/ku
    ▪ Using image registry.k8s.io/metrics-server/metrics-server:v0.9.0
🌟  The 'metrics-server' addon is enabled
metrics-server-768f9f6999-9pc7f    0/1     Running   23 (3m16s ago)   6
mohammedsaad@Saads-MacBook-Air 21-final_project %
```

```
mohammedsaad@Saads-MacBook-Air 21-final_project % kubectl get deployment -n taskboard
kubectl get hpa -n taskboard -o yaml
NAME                       READY   UP-TO-DATE   AVAILABLE   AGE
taskboard-frontend         0/2     2            0           144m
taskboard-postgres         1/1     1            1           144m
taskboard-taskboard-backend   0/2     2            0           144m
apiVersion: v1
items:
- apiVersion: autoscaling/v2
  kind: HorizontalPodAutoscaler
  metadata:
    annotations:
      meta.helm.sh/release-name: taskboard
      meta.helm.sh/release-namespace: taskboard
    creationTimestamp: "2026-10-16T14:32:46Z"
    generation: 1
    labels:
      app.kubernetes.io/managed-by: Helm
    name: taskboard-backend
    namespace: taskboard
    resourceVersion: "204923"
    uid: bb193663-b41d-42e9-a91d-5c9fe9fd2d69
  spec:
    maxReplicas: 6
    metrics:
    - resource:
        name: cpu
        target:
          averageUtilization: 60
          type: Utilization
      type: Resource
    minReplicas: 2
    scaleTargetRef:
      apiVersion: apps/v1
      kind: Deployment
      name: taskboard-taskboard-backend
  status:
    conditions:
    - lastTransitionTime: "2026-10-16T15:47:46Z"
      message: the HPA controller was able to get the target's current scale
      reason: SucceededGetScale
      status: "True"
      type: AbleToScale
    - lastTransitionTime: "2026-10-16T14:32:47Z"
      message: 'the HPA was unable to compute the replica count: failed to get cpu
        utilization: unable to get metrics for resource cpu: unable to fetch metrics
        from resource metrics API: the server is currently unable to handle the request
        (get pods.metrics.k8s.io)'
      reason: FailedGetResourceMetric
      status: "False"
      type: ScalingActive
    currentMetrics:
    - type: ""
    currentReplicas: 2
    desiredReplicas: 0
    observedGeneration: 1
kind: List
metadata:
  resourceVersion: ""
```

```
mohammedsaad@Saads-MacBook-Air 21-final_project % kubectl get hpa -n taskboard -w
NAME                REFERENCE                                  TARGETS              MINPODS   MAXPODS   REPLICAS   AGE
taskboard-backend   Deployment/taskboard-taskboard-backend     cpu: <unknown>/60%   2         6         2          145m
STATUS              REASON                                     MESSAGE
Failure             InternalError                              an error on the server ("unable to decode an event from the watch stream: http2: client connection lost") has prevented the request from succeeding
mohammedsaad@Saads-MacBook-Air 21-final_project %
```

HPA

MONITORING

First show the application metrics endpoint, then Prometheus and Grafana.
curl http://localhost:8000/metrics
kubectl port-forward -n monitoring svc/prometheus 9090:9090

The Prometheus query view proves Prometheus itself is reachable. During the live demo, show the Targets page and confirm the TaskBoard scrape target is UP; then open Grafana and show a panel using TaskBoard HTTP metrics. The captured up query also records failed infrastructure targets, so it should be explained as diagnostic evidence rather than presented as all targets healthy.





2026

```
mohammedsaad@Saads-MacBook-Air 21-final_project % kubectl get pods -A | grep -i prometheus
kubectl get svc -A | grep -i prometheus
monitoring      prometheus-6b5b8798b7-4bkjg                          0/1     Unknown                 14 (4m7s ago)    5d
monitoring      prometheus                           NodePort       10.97.72.188     <none>        9090:30090/TCP              5d
mohammedsaad@Saads-MacBook-Air 21-final_project % kubectl port-forward -n monitoring svc/prometheus 9090:9090
Forwarding from 127.0.0.1:9090 -> 9090
Forwarding from [::1]:9090 -> 9090
Handling connection for 9090
Handling connection for 9090
Handling connection for 9090
Handling connection for 9090
Handling connection for 9090
Handling connection for 9090
```

Prometheus Alerts Graph Status Help

Use local time | Enable query history | Enable autocomplete | Enable highlighting | Enable linter

up

Execute

Load time: 856ms Resolution: 14s Result series: 3

Table Graph

2026-10-16 17:13:54

up{**instance**="kube-state-metrics.monitoring.svc:8080", **job**="kube-state-metrics"}	0
up{**instance**="localhost:9090", **job**="prometheus"}	1
up{**instance**="minikube", **job**="kubernetes-cadvisor"}	0

Remove Panel

Add Panel

```
mohammedsaad@Saads-MacBook-Air 21-final_project % ls troubleshooting
cat troubleshooting/broken-image.yaml
broken-image.yaml       broken-service.yaml
apiVersion: apps/v1
kind: Deployment
metadata: {name: taskboard-broken-image, namespace: taskboard}
spec:
  replicas: 1
  selector: {matchLabels: {app: broken-image}}
  template:
    metadata: {labels: {app: broken-image}}
    spec:
      containers: [{name: backend, image: ghcr.io/example/taskboard-backend:does-not-exist}]
```

FAILURE SIMULATION

2026

Apply one deliberate failure, inspect it, explain the cause, and remove or repair it. The repository includes manifests for a broken image and a broken Service in troubleshooting/.

kubectl apply -f troubleshooting/broken-image.yaml

kubectl get pods -n taskboard

kubectl describe pod -n taskboard <broken-pod-name>

kubectl delete -f troubleshooting/broken-image.yaml

MOHAMMED
SAAD

2026

THANK YOU

For your attention



